

AI-Powered Skill League Engine

What Is the AI-Powered Skill League Engine?

It's an AI system that:

1. Calculates each player's true skill level
2. Places players into dynamic divisions (Bronze, Silver, Gold, Pro)
3. Automatically promotes/demotes players
4. Ensures fair competition
5. Personalizes rewards and matchmaking

Think:

- Ranking system like **League of Legends**
- Competitive tiers like **Valorant**
- ELO-style rating like **World Chess Championship**

But applied to **skill-based fantasy gaming**.

How It Works (Step-by-Step)

Skill Rating Calculation (Core Engine)

Each player gets a dynamic **Skill Score**.

Instead of just:

- Wins
- Total points

We calculate:

AI-Based Skill Formula

Skill Score =

- Win rate (weighted)
- Average rank vs division

- Performance consistency
- Risk-adjusted decision accuracy
- Strength of opponents

You can use:

- ELO algorithm
- Or AI regression model
- Or hybrid approach

2 Dynamic League Placement

Based on Skill Score:

Skill Score League

0–1000 Bronze

1000–1500 Silver

1500–2000 Gold

2000+ Pro

After every contest:

- Score recalculates
- AI checks promotion threshold
- Player moves automatically

🎉 Promotion feels like leveling up in esports.

3 AI Matchmaking

Instead of:

“Everyone competes against everyone”

AI matches:

- Bronze vs Bronze
- Gold vs Gold
- Pro vs Pro

This prevents:

- Beginners losing instantly
- Experts farming easy wins

Reward Optimization

Prize pools adapt by skill tier:

- Bronze: lower entry fee, learning focused
- Silver: medium reward
- Gold: high competitive
- Pro: elite tournaments

AI predicts:

- Drop-off risk
- Retention likelihood
- Engagement level

And adjusts incentives.

What Real-World Problems This Solves

Now the important part for judges:

Problem 1: Beginner Drop-Off

In fantasy gaming:

New users lose against experienced players → they quit.

This is a MAJOR retention problem globally.

Solution:

AI places beginners in Bronze league.

They compete fairly.

They win sometimes.

They stay longer.

 Higher retention = higher revenue.

Problem 2: Unfair Competition

Currently:

- Experts dominate
- Casual players lose money
- Platform looks biased

Your engine ensures:

Fair competition by skill tier.

This increases:

- Trust
- Transparency
- Long-term engagement



Problem 3: Smurfing (Fake Accounts)

High-level players create new accounts to dominate beginners.

AI detects:

- Abnormal win streaks
- Performance spikes
- Decision efficiency

Automatically boosts them to higher league.

Problem solved.



Problem 4: Platform Sustainability

If only top 5% win money:

The rest leave.

Dynamic leagues create:

Multiple mini-ecosystems.

Now:

Top players compete with top players.

Mid-tier players compete in their own economy.

It stabilizes the platform.



Problem 5: Regulatory Concerns

Skill gaming must prove:

“It is based on skill, not luck.”

Your AI system:

- Measures decision quality
- Measures consistency

- Tracks skill growth

This creates:

Documented skill-based proof.

Huge advantage legally.



Why This Is Strategic for Neuball

Neuball = Sri Lanka's first skill-based fantasy platform.

Your system makes them:

“The first AI-tiered skill fantasy ecosystem in South Asia.”

That is INVESTOR language.



MVP Implementation (Realistic for You)

You can build:

Backend

- Node.js server
- Python microservice for skill calculation
- MongoDB for player stats

Core Algorithm

- ELO implementation
- Skill score recalculation after each contest

Frontend

- React dashboard
- League badges
- Promotion animation
- Skill progress bar



How You Should Pitch This

Start like this:

“Current fantasy platforms treat all players equally — but skill is not equal.
Our AI-Powered Skill League Engine creates dynamic, fair, and sustainable competitive ecosystems.”

Then show:

- Skill flow diagram
- League progression
- Business impact



Business Impact

- Higher retention
- Increased trust
- Reduced churn
- Increased average revenue per user
- Strong regulatory positioning
- Differentiation in market

This is not just a feature.

This is platform infrastructure.



Recommended Tech Stack (MVP + Scalable)



1

Frontend (Player Dashboard & League UI)

Framework:

-  React (Vite or Next.js)

UI:

- Tailwind CSS
- Framer Motion (promotion animations)
- Recharts (skill graph visualization)

What it shows:

- Current League (Bronze/Silver/Gold/Pro)
- Skill Score progress bar
- Promotion/Demotion alerts
- Leaderboard by division
- Match history



2 Backend (Core Platform API)

Runtime:

- Node.js (Express.js)

Responsibilities:

- Contest results submission
- Player stats storage
- League management
- API for frontend
- Authentication (JWT)

Why Node?

You're already comfortable with it.



3 AI / Skill Engine (Core Innovation Layer)

Language:

- Python

Libraries:

- NumPy
- Pandas
- Scikit-learn (optional)
- FastAPI (to expose model as API)

Skill Algorithm Options

◆ Option A (Fastest MVP)

Implement:

- ELO rating system

Like:

- Used in chess rankings such as the World Chess Championship

This is simple and powerful.

◆ Option B (More Advanced)

AI regression model:

- Predict expected fantasy performance
- Compare expected vs actual
- Adjust skill score

This looks more “AI impressive” for judges.

Database

Recommended:

- MongoDB (easier for hackathon)
OR
- PostgreSQL (if structured relations needed)

Store:

- Player profiles
- Skill score history
- League tier
- Match history

Authentication

- JWT tokens
- bcrypt for password hashing

Hosting (Free Options)

Frontend:

- Vercel

Backend:

- Render / Railway

AI Microservice:

- Render / Fly.io

Database:

- MongoDB Atlas (Free Tier)



How Everything Connects

React Frontend



Node.js Backend (API Layer)



Python AI Skill Engine (FastAPI)



Database (MongoDB)

Flow:

1. Match ends
2. Backend sends player stats to Python AI service
3. AI recalculates skill score
4. Backend updates league tier
5. Frontend displays promotion/demotion